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Pearson Edexcel
International GCSE

Centre Number

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Candidate Number

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Mathematics A

Level 1/2
Paper 1H

WORKED SOLUTIONS



Higher Tier

Thursday 24 May 2018 – Morning
Time: 2 hours

Paper Reference

4MA1/1H

You must have:

Ruler graduated in centimetres and millimetres, protractor, compasses, pen, HB pencil, eraser, calculator. Tracing paper may be used.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain NO credit.

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►

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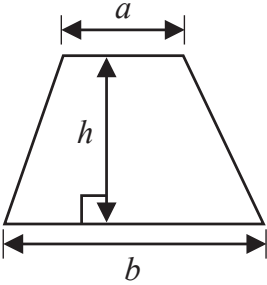
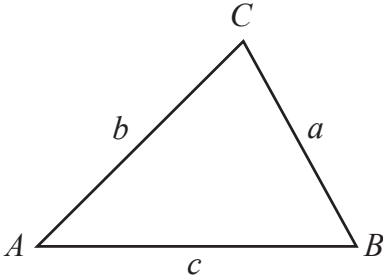
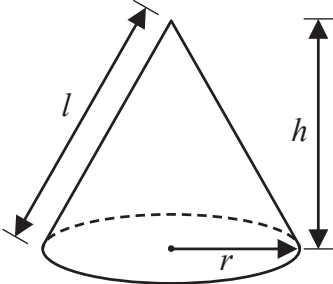
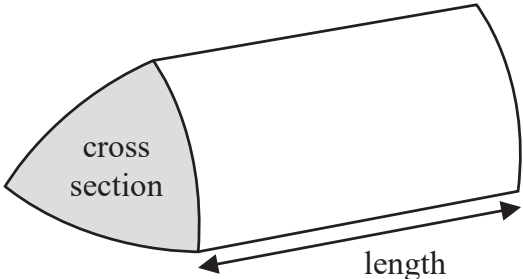
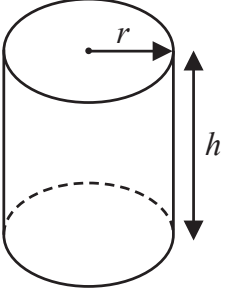
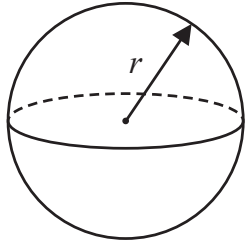


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Pearson

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series Sum to n terms, $S_n = \frac{n}{2} [2a + (n - 1)d]$	Area of trapezium $= \frac{1}{2}(a + b)h$ 
The quadratic equation The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
Trigonometry 	In any triangle ABC Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ Cosine Rule $a^2 = b^2 + c^2 - 2bc \cos A$ Area of triangle $= \frac{1}{2}ab \sin C$
Volume of cone $= \frac{1}{3}\pi r^2 h$ Curved surface area of cone $= \pi r l$ 	Volume of prism $= \text{area of cross section} \times \text{length}$ 
Volume of cylinder $= \pi r^2 h$ Curved surface area of cylinder $= 2\pi r h$ 	Volume of sphere $= \frac{4}{3}\pi r^3$ Surface area of sphere $= 4\pi r^2$ 

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Answer all TWENTY questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 The table shows information about the weights, in kg, of 40 parcels.

Weight of parcel (p kg)	Frequency
$0 < p \leq 1$ $0.5 \times 19 = 9.5$	
$1 < p \leq 2$ $1.5 \times 12 = 18$	
$2 < p \leq 3$ $2.5 \times 5 = 12.5$	
$3 < p \leq 4$ $3.5 \times 2 = 7$	
$4 < p \leq 5$ $4.5 \times 2 = 9$	

40

56

(a) Write down the modal class.

$0 < p \leq 1$
(1)

(b) Work out an estimate for the mean weight of the parcels.

$56 \div 40 = 1.4$

1.4 kg
(4)

(Total for Question 1 is 5 marks)



2 There are some people in a cinema.

$\frac{3}{5}$ of the people in the cinema are children.

For the children in the cinema,

$$\text{number of girls} : \text{number of boys} = 2 : 7$$

There are 170 girls in the cinema.

Work out the number of adults in the cinema.

children
 $\frac{3}{5}$

Girls : Boys
2 : 7

170

$$170 \div 2 \times 7$$

$$595$$

$$\text{children} = 170 + 595$$

$$= 765$$

$$\frac{3}{5} = 765$$

$$\frac{1}{5} = 255$$

$$\text{so } \frac{2}{5} = 510$$

510

(Total for Question 2 is 5 marks)



- 3 (a) Simplify $y^5 \times y^9$

$$y^{5+9}$$

$$y^{14} \quad (1)$$

- (b) Simplify $(2m^3)^4$

$$2^4 m^{3 \times 4} = 16m^{12}$$

$$16m^{12} \quad (2)$$

- (c) Solve $5(x+3) = 3x-4$
Show clear algebraic working.

$$5x + 15 = 3x - 4$$

$$2x = -19$$

$$x = -\frac{19}{2}$$

$$x = -9.5 \quad (3)$$

- (d) (i) Factorise $x^2 + 2x - 24$

$$(x+6)(x-4)$$

$$(x+6)(x-4) \quad (2)$$

- (ii) Hence, solve $x^2 + 2x - 24 = 0$

$$(x+6)(x-4) = 0$$

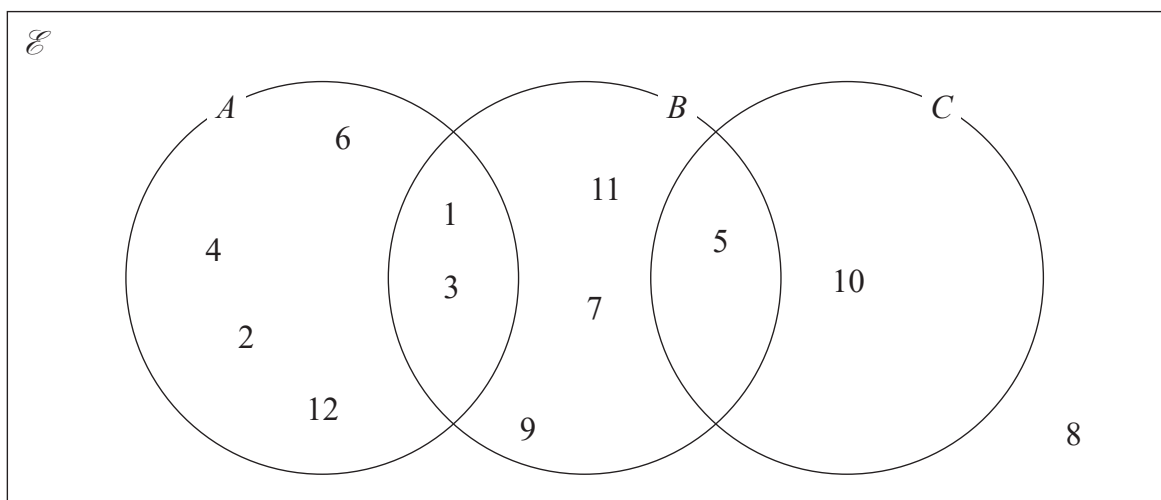
$$x = -6 \quad x = 4$$

$$x = 4, x = -6 \quad (1)$$

(Total for Question 3 is 9 marks)



4 Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

1, 2, 3, 4, 6, 12

(ii) $B \cup C$

1 3 5 7 9 10 11

(2)

Brian writes down the statement $A \cap C = \emptyset$

(b) Is Brian's statement correct?

You must give a reason for your answer.

Yes, there is no intersection between A and C.

(1)

One of the numbers ^{$\nwarrow$ 12} in the Venn diagram is picked at random.

(c) Find the probability that this number is in set C' 10 not in C

$\frac{10}{12}$

(2)

(Total for Question 4 is 5 marks)



- 5 (a) Write 8×10^4 as an ordinary number.

80000

(1)

- (b) Work out $(3.5 \times 10^5) \div (7 \times 10^8)$
Give your answer in standard form.

$$\begin{aligned} & 0.5 \times 10^{5-8} \\ = & 0.5 \times 10^{-3} \\ = & 5 \times 10^{-4} \end{aligned}$$

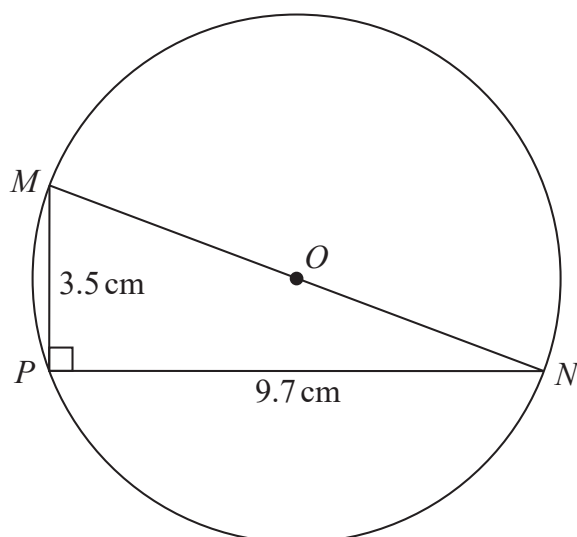
5×10^{-4}

(2)

(Total for Question 5 is 3 marks)



Diagram **NOT**
accurately drawn



M , N and P are points on a circle, centre O .
 MON is a diameter of the circle.

$$MP = 3.5 \text{ cm}$$

$$PN = 9.7 \text{ cm}$$

$$\text{Angle } MPN = 90^\circ$$

Work out the circumference of the circle.

Give your answer correct to 3 significant figures.

$$\begin{aligned} MN^2 &= 3.5^2 + 9.7^2 \\ &= 106.34 \end{aligned}$$

$$MN = \sqrt{106.34} = 10.31..$$

$$\begin{aligned} C &= \pi \times 10.31.. \\ &= 32.39656.. \\ &\quad \uparrow \\ &= 32.4 \text{ (3sf.)} \end{aligned}$$

..... **32.4** cm

(Total for Question 6 is 4 marks)

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- 7 Chao bought a boat for HK\$160 000
The value of the boat depreciates by 4% each year.

0.96

- (a) Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest HK\$.

$$160\,000 \times 0.96^3$$

$$= 141\,557.76$$

↑

HK\$ 141 558
(3)

Jalina gets a salary increase of 5%
Her salary after the increase is HK\$252 000

- (b) Work out Jalina's salary before the increase.

$$105\% = 252\,000$$

$$\div 105 \rightarrow 190 = 2400$$

$$\times 100 \rightarrow 100\% = 240\,000 \times 100$$

HK\$ 240 000
(3)

(Total for Question 7 is 6 marks)



8 $A = 3^5 \times 5 \times 7^3$
 $B = 2^3 \times 3 \times 7^4$

(a) (i) Find the Highest Common Factor (HCF) of A and B .

$$3 \times 7^3$$

$$1029$$

(ii) Find the Lowest Common Multiple (LCM) of A and B .

$$1029 \times 2^3 \times 3^4 \times 5 \times 7$$

$$23\ 337720$$

(2)

$$A = 3^5 \times 5 \times 7^3$$

$$B = 2^3 \times 3 \times 7^4$$

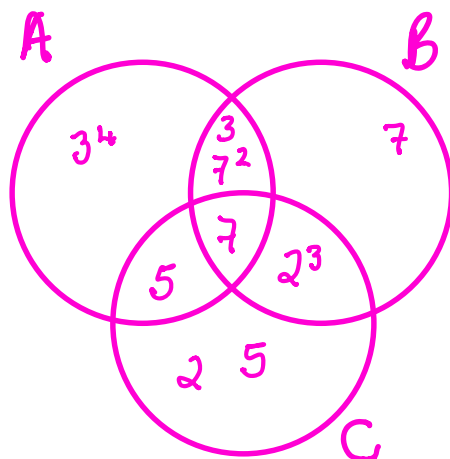
$$C = 2^p \times 5^q \times 7^r$$

Given that

the HCF of B and C is $2^3 \times 7$

the LCM of A and C is $2^4 \times 3^5 \times 5^2 \times 7^3$

(b) find the value of p , the value of q and the value of r .



$$C = 2^4 \times 5^2 \times 7$$

$$p = 4$$

$$q = 2$$

$$r = 1$$

(2)

(Total for Question 8 is 4 marks)



- 9 The diagram shows a right-angled triangle.

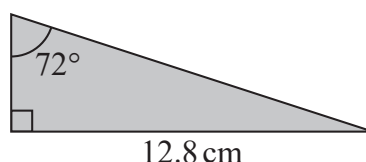


Diagram NOT
accurately drawn

Five of these triangles are put together to make a shape.

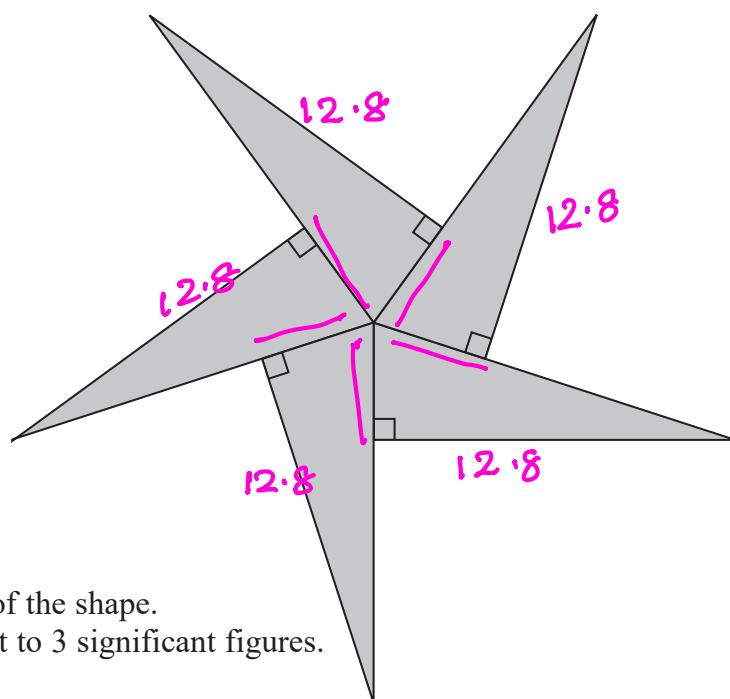
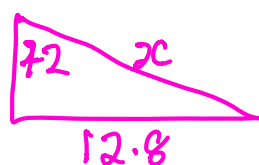


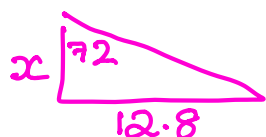
Diagram NOT
accurately drawn

Calculate the perimeter of the shape.
Give your answer correct to 3 significant figures.



$$\sin 72 = \frac{12.8}{x}$$

$$x = \frac{12.8}{\sin 72} = 13.4587...$$



$$\tan 72 = \frac{12.8}{x}$$

$$x = \frac{12.8}{\tan 72} = 4.15897...$$

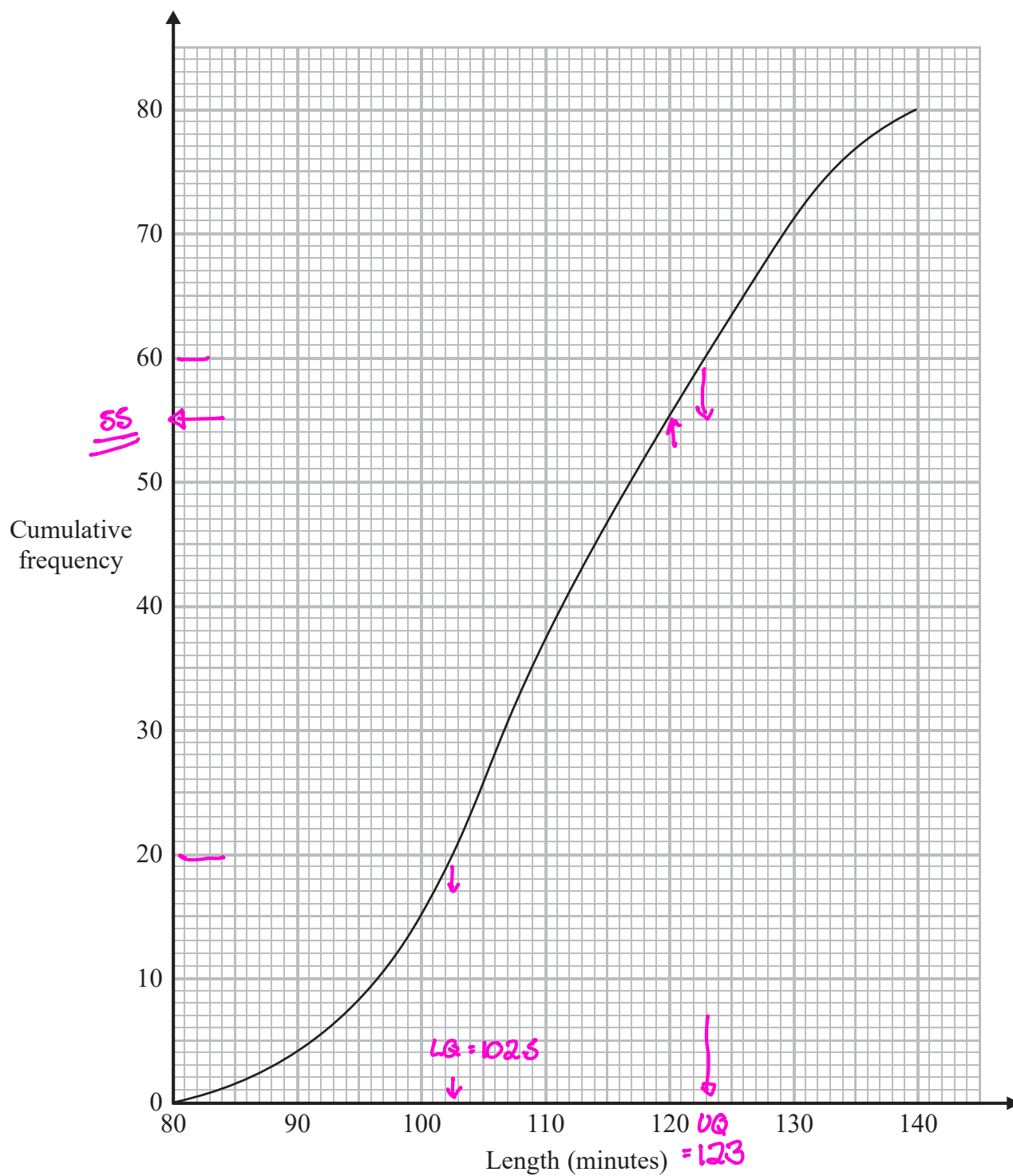
$$\begin{aligned} \text{Perimeter} &= 5 \times 12.8 + 5 \times 13.4587 - 5(4.15897) \\ &= 110.498... \\ &= 110 \text{ (3 s.f.)} \end{aligned}$$

110..... cm

(Total for Question 9 is 5 marks)



- 10 The cumulative frequency graph shows information about the length, in minutes, of each of 80 films.



- (a) Use the graph to find an estimate for the interquartile range.

$$123 - 102.5$$

..... 20.5 minutes
(2)



Clare says,

“More than 35% of these films are over 120 minutes long.”

(b) Is Clare correct?

Give a reason for your answer.

$$80 - 55 = 25 \text{ films}$$

$$\frac{25}{80} \times 100 = 31.25\%$$

No, its approximately 31.25%

(3)

(Total for Question 10 is 5 marks)



11 (a) Expand and simplify $(2x - 1)(x + 3)(x - 5)$

$$(2x - 1)(x + 3) = 2x^2 + 6x - x - 3$$

$$= 2x^2 + 5x - 3$$

$$(2x^2 + 5x - 3)(x - 5)$$

$$= 2x^3 - 10x^2 + 5x^2 - 25x - 3x + 15$$

$$= 2x^3 - 5x^2 - 28x + 15$$

$$2x^3 - 5x^2 - 28x + 15$$

(3)

(b) Solve $3x^2 + 6x - 5 = 0$

Show your working clearly.

Give your solutions correct to 3 significant figures.

$$a = 3 \quad b = 6 \quad c = -5$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4 \times 3 \times -5}}{2 \times 3}$$

$$= \frac{-6 \pm \sqrt{36 + 60}}{6}$$

$$= \frac{-6 \pm \sqrt{96}}{6}$$

$$0.63299... \quad -2.63299...$$

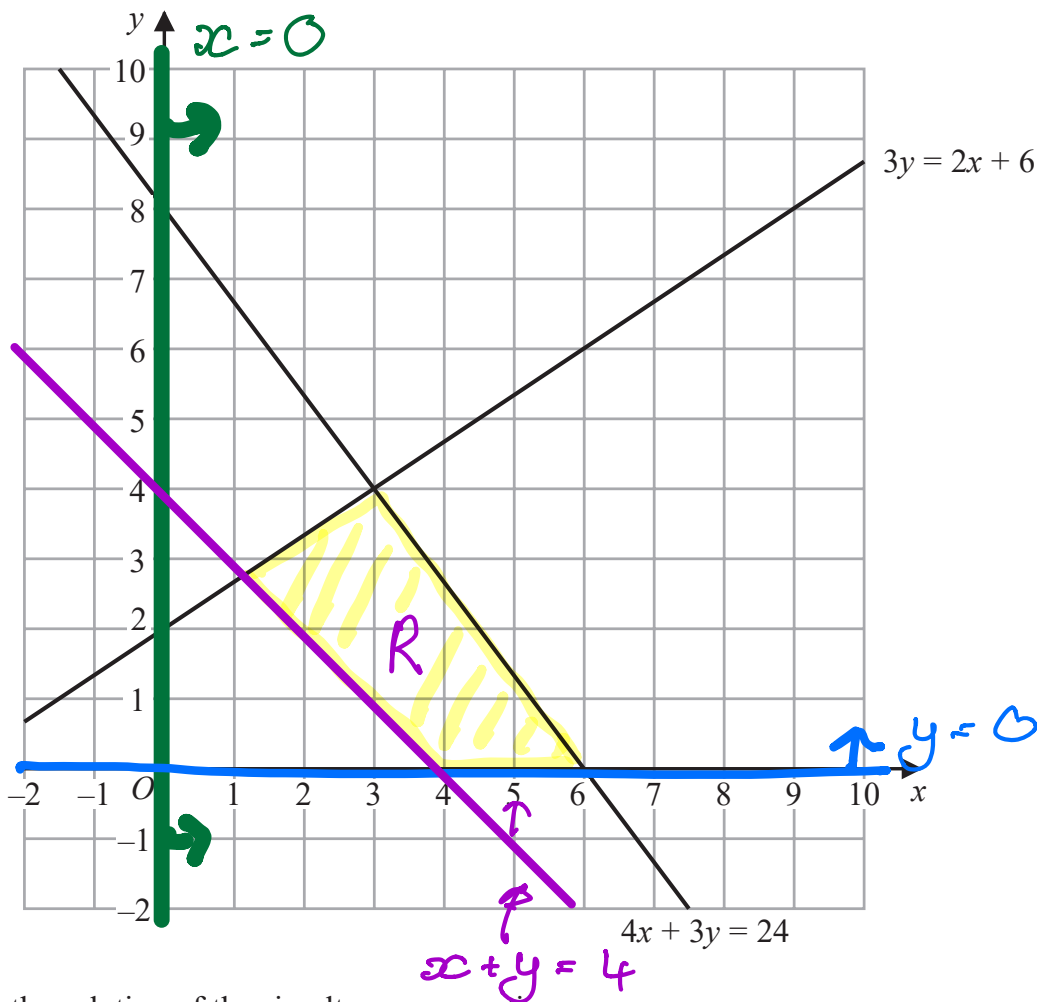
$$0.633, -2.63$$

(3)

(Total for Question 11 is 6 marks)



12 The diagram shows two straight lines drawn on a grid.



(a) Write down the solution of the simultaneous equations

$$\begin{aligned} 3y &= 2x + 6 \\ 4x + 3y &= 24 \end{aligned}$$

$x = \dots\dots\dots 3$
 $y = \dots\dots\dots 4$
(1)

(b) Show, by shading on the grid, the region defined by all five of the inequalities

$x \geq 0$ $y \geq 0$ $x + y \geq 4$ $3y \leq 2x + 6$ $4x + 3y \leq 24$

Label the region R.

(3)

(Total for Question 12 is 4 marks)

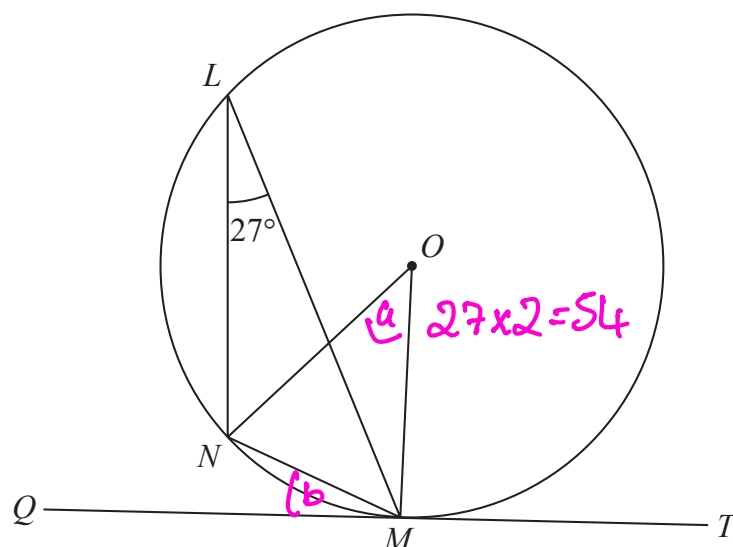


Diagram **NOT**
accurately drawn

L , M and N are points on a circle, centre O .
 QMT is the tangent to the circle at M .

- (a) (i) Find the size of angle NOM .

54

- (ii) Give a reason for your answer.

The angle at the centre is twice the angle
at the circumference

(2)

- (b) (i) Find the size of angle NMQ .

27

- (ii) Give a reason for your answer.

alternate segment theorem

(2)

(Total for Question 13 is 4 marks)



14 The function f is such that

$$f(x) = \frac{3x - 5}{4}$$

(a) Find $f(-7)$

$$f(-7) = \frac{3(-7) - 5}{4} = \frac{-21 - 5}{4}$$

$$\frac{-6 \cdot 5}{4}$$

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$y = \frac{3x - 5}{4}$$

$$4y + 5 = 3x$$

$$x = \frac{4y + 5}{3}$$

$$\text{so } f^{-1}(x) = \frac{4x + 5}{3}$$

$$f^{-1}(x) = \frac{4x + 5}{3}$$

(2)

The function g is such that

$$g(x) = \sqrt{19 - x}$$

(c) Find $fg(3)$

$$g(3) = \sqrt{19 - 3} = \sqrt{16} = 4$$

$$fg(3) = \frac{3 \times 4 - 5}{4} = \frac{7}{4}$$

$$\frac{1 \cdot 7 \cdot 5}{4}$$

(2)

(d) Which values of x cannot be included in any domain of g ?

$$x > 19$$

(2)

(Total for Question 14 is 7 marks)



15 (a) Simplify fully $\left(\frac{256x^{20}}{y^8}\right)^{-\frac{1}{4}}$

$$\left(\frac{y^8}{256x^{20}}\right)^{\frac{1}{4}} = \frac{y^{8 \times \frac{1}{4}}}{\sqrt[4]{256} x^{20 \times \frac{1}{4}}}$$

$$= \frac{y^2}{4x^5}$$

$$\frac{y^2}{4x^5}$$

(2)

(b) Express $\frac{1}{9x^2 - 25} - \frac{1}{6x + 10}$ as a single fraction in its simplest form.

$$9x^2 - 25 \rightarrow (3x - 5)(3x + 5)$$

$$6x + 10 \rightarrow 2(3x + 5)$$

$$\frac{2 - 1(3x - 5)}{2(3x + 5)(3x - 5)}$$

$$= \frac{2 - 3x + 5}{2(3x + 5)(3x - 5)}$$

$$= \frac{7 - 3x}{2(3x + 5)(3x - 5)}$$

$$\frac{7 - 3x}{2(3x + 5)(3x - 5)}$$

(3)

(Total for Question 15 is 5 marks)



- 16 A frustum is made by removing a small cone from a large cone.
The cones are mathematically similar.

$$Vol = \frac{1}{3} \pi r^2 h$$

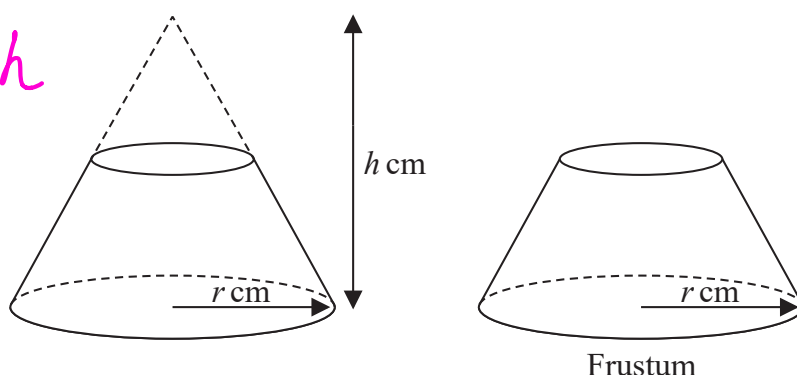


Diagram **NOT**
accurately drawn

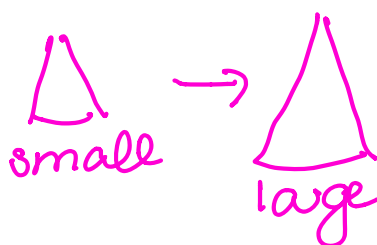
The large cone has base radius r cm and height h cm.

Given that

$$\frac{\text{volume of frustum}}{\text{volume of large cone}} = \frac{98}{125}$$

$$\begin{aligned} \text{Volume of small cone} &= 125 - 98 \\ &= 27 \end{aligned}$$

find an expression, in terms of h , for the height of the frustum.



$$\begin{aligned} \text{VOLUME} \\ \text{scale factor} &= \frac{125}{27} \end{aligned}$$

$$\text{length SF} = \frac{5}{3}$$

$$\begin{aligned} &\left[\begin{array}{l} \rightarrow 3 \\ \text{FRUSTUM} \end{array} \right] \left[\begin{array}{l} h \rightarrow 5 \end{array} \right] \end{aligned}$$

$$\text{height of frustum} = \frac{2}{5}h$$

$$\frac{2}{5}h \dots \text{cm}$$

(Total for Question 16 is 4 marks)



17 The diagram shows parallelogram $ABCD$.

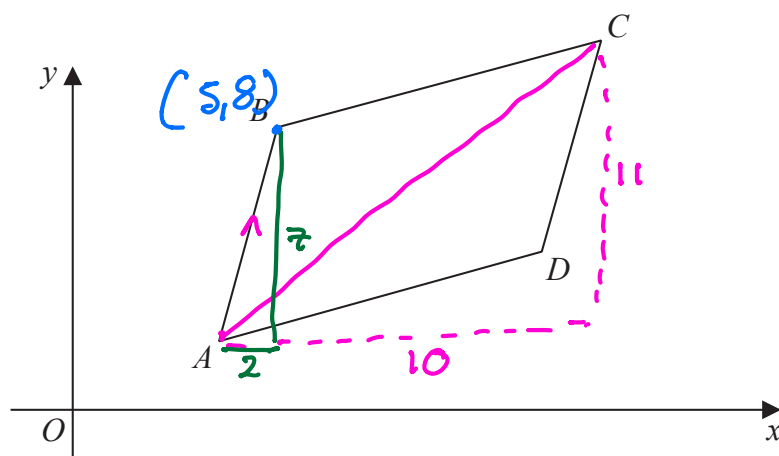


Diagram **NOT** accurately drawn

$$\vec{AB} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad \vec{AC} = \begin{pmatrix} 10 \\ 11 \end{pmatrix}$$

The point B has coordinates $(5, 8)$ so $A = (3, 1)$

(a) Work out the coordinates of the point C .

$$C = 13, 12$$

$$\left(\frac{13}{3}, \frac{12}{3} \right)$$

The point E has coordinates $(63, 211)$

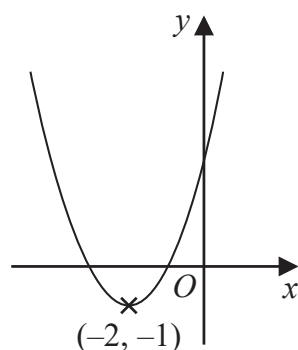
(b) Use a vector method to prove that ABE is a straight line.

$$\begin{aligned} E &= (63, 211) & AE &= \begin{pmatrix} 60 \\ 210 \end{pmatrix} & AB &= \begin{pmatrix} 2 \\ 7 \end{pmatrix} \\ B &= (5, 8) \\ A &= (3, 1) \\ \therefore AE &= 30AB \\ 30 \times 2 &= 60 & \therefore ABE &\text{ is a straight line} \\ 30 \times 7 &= 210 \end{aligned}$$

(2)

(Total for Question 17 is 5 marks)





The diagram shows the curve with equation $y = f(x)$

The coordinates of the minimum point of the curve are $(-2, -1)$

(a) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(x - 5)$

$\rightarrow 5$

(3, -1)

(ii) $y = \frac{1}{2}f(x)$

(-2, -0.5)
(2)

The graph of $y = a \sin(x - b)^\circ + c$ for $-90 \leq x \leq 450$ is drawn on the grid below.



(b) Find the value of a , the value of b and the value of c .

$2 \sin(x - 90) + 1$

$a =$ 2

$b =$ 90

$c =$ 1

(3)

(Total for Question 18 is 5 marks)



19 Jack plays a game with two fair spinners, A and B.

Spinner A can land on the number 2 or 3 or 5 or 7

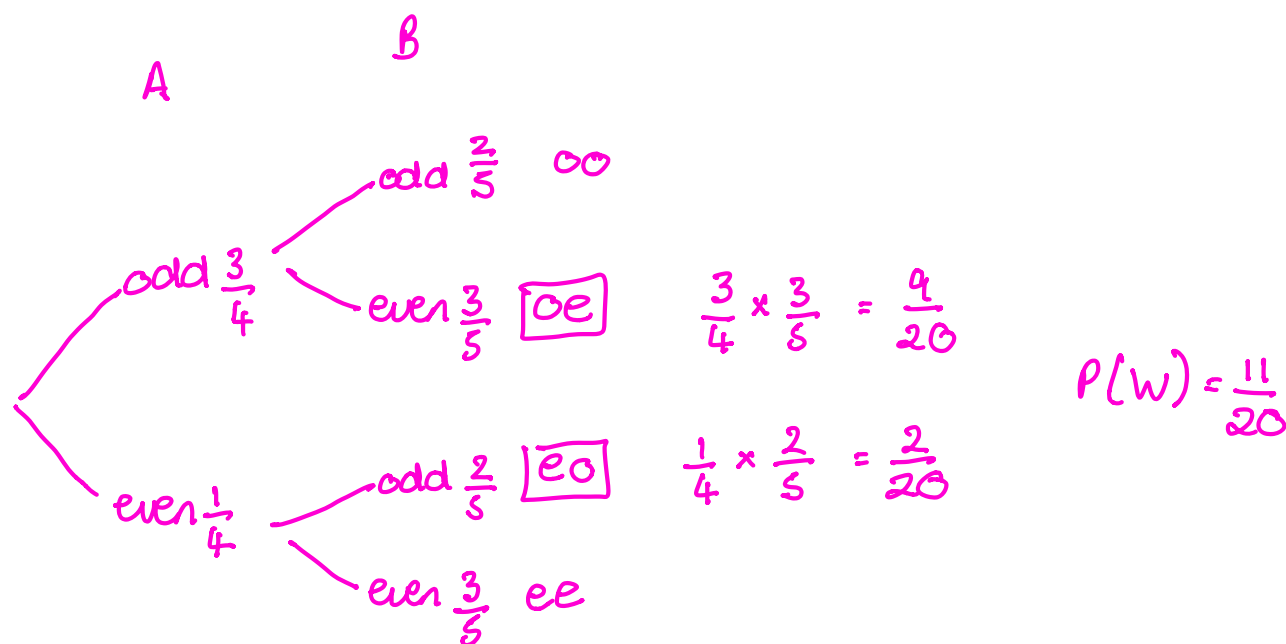
Spinner B can land on the number 2 or 3 or 4 or 5 or 6

Jack spins both spinners.

He wins the game if one spinner lands on an odd number **and** the other spinner lands on an even number.

Jack plays the game twice.

Work out the probability that Jack wins the game both times.



$$P(W \text{ for 2 games}) = \frac{11}{20} \times \frac{11}{20}$$

$$= \frac{121}{400}$$

$$\frac{121}{400}$$

(Total for Question 19 is 4 marks)



20 ABC is an isosceles triangle such that

$$AB = AC$$

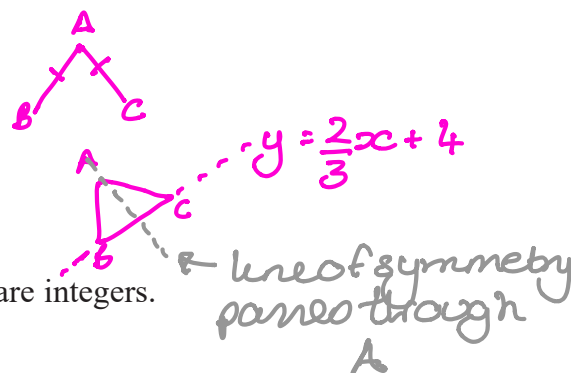
A has coordinates $(4, 37)$

B and C lie on the line with equation $3y = 2x + 12$

Find an equation of the line of symmetry of triangle ABC .

Give your answer in the form $px + qy = r$ where p, q and r are integers.

Show clear algebraic working.



$$3y = 2x + 12$$

$$y = \frac{2}{3}x + 4$$

$$\text{gradient of } BC = \frac{2}{3}$$

$$\therefore \text{gradient of perp. line} = -\frac{3}{2}$$

Perpendicular line

$$y = -\frac{3}{2}x + c$$

$$A(4, 37)$$

$x \quad y$

$$37 = -\frac{3}{2} \times 4 + c$$

$$37 + 6 = c$$

$$c = 43$$

$$\text{so } y = -\frac{3}{2}x + 43$$

$$\times 2 \quad 2y = -3x + 86$$

$$3x + 2y = 86$$

$$3x + 2y = 86$$

(Total for Question 20 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS

